Arithmetic obstructions in positive determinant approximations to $\beta(4)$

Research note prepared with Codex

31 October 2026

Abstract

We study two explicit positive determinant constructions for the Dirichlet beta value $\beta(4)$. Their real determinants become small, but their primitive integer polynomials have values that grow. For a cubic-tail kernel, we prove a uniform two-sided comparison with a Cauchy determinant and show that every sequence of consecutive principal offsets has primitive-value lower exponent at least $2 - 2\log 2$ on the quadratic scale. For a difference-variable Gram construction, we determine the exact primitive multiplier valuation at every prime $p > 4n$ and obtain the lower exponent $6 - \log 4$. The arithmetic bounds apply after removal of every common coefficient factor. They therefore exclude these specified determinant sequences as sources of small primitive values, without excluding other weights, subspaces or determinant constructions. All arguments particular to the two families are given in the paper.

1 Introduction

Put

$$\theta = \beta(4) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^4}. \tag{1.1}$$

Positive integral representations provide nonzero quantities involving a constant, and their determinants can decay exponentially in the square of the matrix size. Such real decay is useful in an irrationality argument only if the necessary arithmetic normalization is sufficiently inexpensive. For a rational polynomial Q, clearing denominators and then removing the common factor of its integer coefficients gives its *primitive integer representative*. This paper proves that this final representative, not just a particular denominator-clearing multiple, has a growing value at θ in two determinant families.

The relation between positive moments, determinant decay and irrationality criteria is developed systematically by Zudilin [6]. One motivation for studying $\beta(4)$ determinants is the mixed determinant strategy for Catalan's constant described by OpenAI [4, Section 1.2], which treats nonvanishing, the real estimate and prime valuations separately. The positive families considered here examine only part of the much wider space of determinant designs. Prévost [5] constructs positive-weight representations and Padé approximants for Hurwitz zeta values. Rational modifications of moment functionals belong to the classical orthogonal-polynomial framework; Krattenthaler [3] gives determinant formulas that include such modifications and repeated parameters. More directly, Carreira [2] investigates determinant constructions for $\beta(4)$, including positive functionals, primitive polynomial calculations and the competing real and arithmetic costs. Brown's discussion [1, Section 7.4] of denominator clearing, common divisors and congruences further emphasizes that the full coefficient

content must be accounted for. We use these works to situate the question; no claimed global irrationality result in a recent manuscript is an input to our proofs.

The contributions here are specific uniform estimates and exact local arithmetic for the two families defined below. The first combines a two-sided Cauchy comparison with a coefficient whose prime valuation is forced by a unique surviving principal minor. The second has a full-rank coefficient of θ and a leading residue block whose determinant determines the primitive multiplier exactly at large primes.

1.1 The two families and main results

For integers $a \geq 0$ and $\ell \geq 2$, write

$$S_\ell(a) = \sum_{j=0}^{a-1} \left(\frac{1}{(4j+1)^\ell} - \frac{1}{(4j+3)^\ell} \right), \qquad S_\ell(0) = 0. \tag{1.2}$$

For distinct positive integer nodes $a_1, \ldots, a_n$, let $R(a)$ be the rational symmetric matrix

$$R(a)_{ii} = -S_4(a_i) + \frac{S_3(a_i)}{12a_i}, \qquad R(a)_{ij} = \frac{a_j S_3(a_j) - a_i S_3(a_i)}{6(a_j^2 - a_i^2)} \quad (i \neq j). \tag{1.3}$$

For consecutive nodes $a_i = m + i$, $1 \leq i \leq n$, set

$$Q_{n,m}(X) = \det(XI_n + R(a)), \qquad Z_{n,m}(X) = H_{n,m} Q_{n,m}(X) = \mathrm{Prim}(Q_{n,m})(X), \tag{1.4}$$

where $H_{n,m} > 0$ and $Z_{n,m} \in \mathbb{Z}[X]$ has coefficient gcd one.

Theorem 1.1 (Consecutive principal determinants)**.** *For every sequence of nonnegative integer offsets m_n, $Q_{n,m_n}(\theta) > 0$ and*

$$\liminf_{n\to\infty} \frac{\log Z_{n,m_n}(\theta)}{n^2} \geq 2 - 2\log 2 > 0. \tag{1.5}$$

The theorem allows arbitrary growth of m_n, including $m_n/n \to \infty$. The underlying kernel admits a uniform comparison for arbitrary positive integer row and column nodes (Theorem 3.3); the arithmetic conclusion above is restricted to consecutive principal nodes.

For the second family, define

$$T_\ell(j) = \sum_{k=0}^{j-1} \frac{(-1)^k}{(2k+1)^\ell}, \qquad C_m = \binom{2m}{m}, \tag{1.6}$$

$$B_m = -C_m T_4(m) + \frac{1}{6} \sum_{\substack{0 \leq j \leq 2m \\ j \neq m}} \binom{2m}{j} \frac{T_3(j)}{j-m}, \tag{1.7}$$

$$Q_n^\Delta(X) = \det[C_{i+j}X + B_{i+j}]_{0 \leq i,j < n}, \qquad Z_n^\Delta = \kappa_n Q_n^\Delta = \mathrm{Prim}(Q_n^\Delta), \quad \kappa_n > 0. \tag{1.8}$$

Empty sums are zero. In particular $B_0 = 0$ and $B_1 = -149/81$.

Theorem 1.2 (Difference-variable Gram determinants)**.** *The polynomial Q_n^Δ has degree n and leading coefficient 2^{n-1}, and $Q_n^\Delta(\theta) > 0$. For every prime $p > 4n$,*

$$v_p(\kappa_n) = 3 \max\left\{0,\, 2n - 2 - \left\lfloor \frac{p}{4} \right\rfloor \right\}. \tag{1.9}$$

Moreover,

$$\log Z_n^\Delta(\theta) \geq (6 - \log 4) n^2 + o(n^2). \tag{1.10}$$

Both theorems imply that the corresponding primitive values eventually exceed one and that their nth roots tend to infinity. They exclude the specified sequences as small primitive values; they are not statements against all determinant methods for θ.

1.2 Proof organization and background

Section 2 fixes primitive normalization and the degree-sensitive small-value criterion. Section 3 derives the positive cubic-tail kernel and its Cauchy comparison. Section 4 proves Theorem 1.1. Sections 5 and 6 construct the positive Gram measure and prove Theorem 1.2. Section 7 states the scope and verification status. Appendix A treats the additional zero node.

We use elementary complex integration, linear algebra, Stirling's formula and the prime number theorem in the form

$$\vartheta(x) := \sum_{p \le x} \log p = x + o(x). \tag{1.11}$$

The latter is the only theorem on prime distribution used. The specific kernel identities, determinant comparisons, moment formulas and local valuation arguments are proved below. No numerical optimization or finite enumeration is an input to either asymptotic obstruction.

2 Primitive normalization and small polynomial values

Definition 2.1. For $0 \neq Q \in \mathbb{Q}[X]$, its primitive representative $\mathrm{Prim}(Q)$ is the unique positive rational multiple $\kappa Q \in \mathbb{Z}[X]$ whose coefficients have gcd one. This convention preserves the sign of Q.

For a nonzero rational number, v_p denotes its usual p-adic valuation, normalized by $v_p(p) = 1$; set $v_p(0) = +\infty$. Saying that a rational number is p-integral means that its denominator in lowest terms is not divisible by p. Reduction modulo p below is always performed on p-integral quantities.

Lemma 2.2. *If $Q(X) = \sum_j q_j X^j \neq 0$ and $\mathrm{Prim}(Q) = \kappa Q$, then*

$$v_p(\kappa) = -\min_j v_p(q_j) \tag{2.1}$$

for every prime p. If Q is monic, κ is a positive integer. If the leading coefficient of Q is 2^e with $e \ge 0$, then $v_p(\kappa) \ge 0$ for odd p and $v_2(\kappa) \ge -e$.

Proof. Integral primitive coefficients have minimum valuation zero at every prime, which proves (2.1). A monic polynomial has a coefficient of valuation zero at every prime. A leading coefficient 2^e has valuation zero at odd primes and valuation e at 2. □

A sufficient common denominator can overstate the primitive multiplier. For example, $Q(X) = (6X + 12)/5$ becomes integral after multiplication by 5, but $\mathrm{Prim}(Q) = X + 2$ and its primitive multiplier is $5/6$. A lower bound on a selected denominator-clearing integer therefore need not be a lower bound on the primitive multiplier. Formula (2.1) is the distinction used in both main proofs.

If U, V are invertible rational matrices, then $\det(UM(X)V) = \det U \det V \det M(X)$. Consequently the primitive polynomial of a full pencil is unchanged up to sign by rational invertible row and column operations. Passing to a different subspace or a different pencil is a different construction and is not covered by this observation.

For completeness, if $\theta = a/b \in \mathbb{Q}$ with $b \ge 1$ and $Z \in \mathbb{Z}[X]$ has degree at most n, then $b^n Z(\theta) \in \mathbb{Z}$. Thus nonzero values with $|Z_n(\theta)|^{1/n} \to 0$ would contradict rationality. The degree factor is essential:

merely having $Z_n(\theta) \to 0$ with increasing degree need not do so. For example, $Z_n(X) = X^n$ has value 2^{-n} at the rational number $1/2$. The main theorems give growth in the opposite direction for their two specified families.

3 The positive cubic-tail kernel

3.1 A direct integral representation

For $\operatorname{Re} z \geq 0$, define

$$f(z) = \sum_{j=0}^{\infty} \left((4j+1+4z)^{-3} - (4j+3+4z)^{-3} \right). \tag{3.1}$$

Expanding $\operatorname{sech} u = 2\sum_{k\geq 0}(-1)^k e^{-(2k+1)u}$ and integrating gives

$$f(z) = \frac{1}{4}\int_0^{\infty} u^2 \operatorname{sech}(u) e^{-4zu}\,\mathrm{d}u, \quad f(a) = f(0) - S_3(a), \quad f'(a) = -12(\theta - S_4(a)) \tag{3.2}$$

at nonnegative integer a. The integrated series is absolutely summable, which justifies the expansion and differentiation used here.

We recall two elementary transforms, including their proofs to fix the normalization:

$$\int_0^{\infty} \operatorname{sech}(u)\cos(tu)\,\mathrm{d}u = \frac{\pi}{2}\operatorname{sech}\frac{\pi t}{2}, \tag{3.3}$$

$$\int_0^{\infty} \frac{x(1-\cos(ts))}{x+s^2}\,\mathrm{d}s = \frac{\pi\sqrt{x}}{2}(1-e^{-t\sqrt{x}}), \qquad x>0,\ t\geq 0. \tag{3.4}$$

For (3.3), integrate $e^{itz}/\cosh z$ around the rectangle with vertices $\pm R, \pm R + i\pi$. The top edge contributes $e^{-\pi t}$ times the bottom edge, and the pole at $i\pi/2$ has residue $e^{-\pi t/2}/i$. The vertical edges tend to zero. Hence the full-line transform is $\pi\operatorname{sech}(\pi t/2)$; symmetry gives the displayed formula. For (3.4), the upper-half-plane pole at $i\sqrt{x}$ gives $\int_0^{\infty}\cos(ts)/(s^2+x)\,\mathrm{d}s = \pi e^{-t\sqrt{x}}/(2\sqrt{x})$ for $t>0$, and the value at $t=0$ follows by continuity. Subtract the latter transforms and multiply by x.

Differentiating (3.3) twice under the integral gives

$$f(0) = \frac{\pi^3}{32}, \qquad \operatorname{Re} f(is) = \frac{\pi^3}{32}A(s)(2A(s)^2-1), \quad A(s) = \operatorname{sech}(2\pi s). \tag{3.5}$$

Define

$$g(x) = \frac{\sqrt{x}}{6}(f(0) - f(\sqrt{x})), \quad g(0) = 0, \qquad K(x,y) = \frac{g(y)-g(x)}{y-x}, \tag{3.6}$$

with the derivative value when $x = y$.

Proposition 3.1. *For $x, y \geq 0$,*

$$K(x,y) = \int_0^{\infty} \frac{\omega(s)}{(x+s^2)(y+s^2)}\,\mathrm{d}s, \quad \omega(s) = \frac{\pi^2}{96}s^2 q(s), \quad q(s) = 1 + A(s) - 2A(s)^3. \tag{3.7}$$

The density ω is strictly positive for $s>0$. On squared integer nodes the kernel equals

$$K(a^2,b^2) = \frac{bS_3(b) - aS_3(a)}{6(b^2-a^2)} \qquad (a\neq b), \tag{3.8}$$

$$K(a^2,a^2) = \theta - S_4(a) + \frac{S_3(a)}{12a} \qquad (a>0), \tag{3.9}$$

$$K(0,0) = 2\theta. \tag{3.10}$$

Every square minor with strictly increasing nonnegative row nodes and strictly increasing nonnegative column nodes is positive.

Proof. Use (3.4) with $t = 4u$ in (3.2). The integrands before interchange are nonnegative, so Tonelli's theorem gives

$$g(x) = \int_0^\infty \frac{x}{x+s^2}\sigma(s)\,\mathrm{d}s, \qquad \sigma(s) = \frac{f(0) - \operatorname{Re} f(is)}{3\pi} = \frac{\pi^2}{96}q(s). \tag{3.11}$$

Taking a divided difference yields (3.7). Here

$$q = (1-A)(2A^2+2A+1) > 0 \qquad (s>0). \tag{3.12}$$

Since $\omega(s) = O(s^4)$ at zero and $O(s^2)$ at infinity, the integral is finite even when $x = y = 0$. Differentiation or dominated convergence justifies the diagonal and zero limits. The identities (3.8)–(3.9) follow from (3.2); at zero, $g(x) = -f'(0)x/6 + O(x^{3/2}) = 2\theta x + O(x^{3/2})$.

For increasing nodes x_i, y_j, expanding the two determinants and integrating termwise gives the determinant integration identity

$$\det[K(x_i,y_j)]_{i,j=1}^n = \frac{1}{n!}\int_{(0,\infty)^n} \det\left[\frac{1}{x_i+s_\ell^2}\right]_{i,\ell} \det\left[\frac{1}{y_j+s_\ell^2}\right]_{j,\ell} \prod_{\ell=1}^n \omega(s_\ell)\,\mathrm{d}s_1\cdots\mathrm{d}s_n. \tag{3.13}$$

Absolute integrability follows by expanding into finitely many products of the finite one-variable integrals. This is the usual Andréief–Heine identity used in positive determinant arguments [6, Section 2], with its elementary expansion proof here. For $0 < s_1 < \cdots < s_n$, both Cauchy determinants are positive by the Cauchy product formula. Their product is unchanged by permuting the s_ℓ and is strictly positive away from coincidences. This proves the last assertion. It also proves that increasing the density increases these determinants, even for different row and column nodes. □

Remark 3.2. The density in (3.7) satisfies $\omega(s) \sim \pi^2 s^2/96$ at infinity; it has infinite total mass, but the two resolvent factors make every kernel entry integrable. Carreira's functional [2, Definition 24(B) and Lemma 25] uses a positive density obtained from derivatives of $1/\sinh(\pi s)$, with exponential decay and finite polynomial moments. The representation used here is derived directly, and assumes no polynomial moments of $\omega(s)\,\mathrm{d}s$.

3.2 A uniform comparison with Cauchy determinants

For positive nodes a_i, b_j, write

$$C(a,b)_{ij} = \frac{1}{a_i+b_j}, \qquad \det C(a,b) = \frac{\prod_{i<j}(a_j-a_i)(b_j-b_i)}{\prod_{i,j}(a_i+b_j)}. \tag{3.14}$$

The product formula follows, for example, by multiplying the determinant by its displayed denominator: the numerator is alternating separately in the two node sets, hence contains their Vandermonde factors, and comparison of degrees and one residue fixes the remaining constant to one. Set

$$c_0 = \frac{\pi^3}{192}, \qquad M = 1 + \frac{2}{3\sqrt{6}}, \qquad \tau_0 = \frac{1}{\sqrt{2\pi}}. \tag{3.15}$$

Theorem 3.3 (Uniform Cauchy comparison)**.** *For any two strictly increasing lists of positive integers a_i, b_j,*

$$\left(\frac{c_0}{(1+\tau_0)^2}\right)^n \det C(a,b) \le \det[K(a_i^2, b_j^2)] \le (Mc_0)^n \det C(a,b). \tag{3.16}$$

In particular,

$$\log\det[K(a_i^2, b_j^2)] = \log\det C(a,b) + O(n), \tag{3.17}$$

with an absolute constant independent of all node positions and gaps.

Proof. The maximum of $1 + A - 2A^3$ for $0 \le A \le 1$ is M. Also $\cosh v \ge 1 + v^2/2$ and (3.12) imply

$$\frac{s^2}{s^2+\tau_0^2} \le 1 - \operatorname{sech}(2\pi s) \le q(s) \le M. \tag{3.18}$$

The elementary rational integrals needed for the comparison are

$$\int_0^\infty \frac{s^2\,\mathrm{d}s}{(s^2+a^2)(s^2+b^2)} = \frac{\pi}{2(a+b)}, \tag{3.19}$$

$$\int_0^\infty \frac{s^4\,\mathrm{d}s}{(s^2+\tau^2)(s^2+a^2)(s^2+b^2)} = \frac{\pi}{2} R_\tau(a,b), \tag{3.20}$$

$$R_\tau(a,b) = \frac{ab + \tau(a+b)}{(a+b)(a+\tau)(b+\tau)}. \tag{3.21}$$

To check the second formula, first divide $s^4/((s^2+a^2)(s^2+b^2))$ into $[a^2s^2/(s^2+a^2) - b^2s^2/(s^2+b^2)]/(a^2-b^2)$, and then use the first formula with the extra factor $s^2+\tau^2$. Coincident parameters follow by continuity. The first formula follows by partial fractions and $\int_0^\infty (s^2+a^2)^{-1}\,\mathrm{d}s = \pi/(2a)$.

Applying the positive determinant integral (3.13) to the density bounds gives

$$c_0^n \det[R_{\tau_0}(a_i,b_j)] \le \det[K(a_i^2,b_j^2)] \le (Mc_0)^n \det C(a,b). \tag{3.22}$$

This uses the sign of the determinant integrand; an entrywise matrix comparison alone would not suffice.

We now evaluate the regularized determinant exactly:

$$\begin{aligned}\det[R_\tau(a_i,b_j)] = {}& \det C(a,b) \prod_i \frac{a_i}{a_i+\tau} \prod_j \frac{b_j}{b_j+\tau} \\ & \times \left(1 + \tau\sum_i \frac{1}{a_i} + \tau \sum_j \frac{1}{b_j}\right).\end{aligned} \tag{3.23}$$

Let $u_i = 1/a_i$, $v_j = 1/b_j$. Factoring rows and columns in (3.21) leaves $C + \tau uv^t$. If $w = C^{-1}u$ and $F(x) = \sum_j w_j/(x+b_j)$, then $F(a_i) = 1/a_i$. Therefore

$$\frac{1}{x} - F(x) = \frac{1}{x}\,\frac{\prod_i (1 - x/a_i)}{\prod_j (1 + x/b_j)}.$$

Indeed, after multiplication by $x\prod_j(x+b_j)$, the numerator has degree at most n and vanishes at every a_i; its factor is fixed by the residue one at zero. Comparing constant terms at zero gives $v^tC^{-1}u = F(0) = \sum_i 1/a_i + \sum_j 1/b_j$. The rank-one determinant identity proves (3.23). Finally each integer node is at least one, so the products are at least $(1+\tau_0)^{-2n}$ and the last factor is at least one. This proves (3.16) and (3.17). □

4 The consecutive principal obstruction

4.1 A coefficient with forced valuation

For an odd prime p put $\chi_4(p) = (-1)^{(p-1)/2}$.

Lemma 4.1. *Let $a_1 < \cdots < a_n$ be positive integers with maximum M_0, and set $Q(X) = \det(XI_n + R(a))$. If $p \geq 5$ and $2M_0 < p < 4M_0$, put $r_p = \#\{i : 4a_i > p\}$. Then*

$$v_p([X^{n-r_p}]Q) = -4r_p, \qquad v_p(\kappa(Q)) \geq 4r_p, \tag{4.1}$$

where $\mathrm{Prim}(Q) = \kappa(Q)Q$.

Proof. The quantities a_i, $a_i + a_j$ and $a_i - a_j$ for $i \neq j$ are p-units. Among odd denominators below $4M_0$, the only possible multiple of p is p itself, since $3p > 4M_0$. It appears in $S_\ell(a_i)$ exactly when $4a_i > p$, with sign $\chi_4(p)$. Consequently $p^4R(a)$ is p-integral and its reduction is diagonal:

$$p^4R(a)_{ij} \equiv \begin{cases} -\chi_4(p), & i = j,\ 4a_i > p, \\ 0, & \text{otherwise} \end{cases} \pmod{p}. \tag{4.2}$$

The coefficient of X^{n-r_p} is the sum of all principal minors of R of size r_p. After multiplication by p^{4r_p}, exactly one such minor survives modulo p: that using all the high nodes. Its residue is $(-\chi_4(p))^{r_p} \neq 0$. For $r_p = 0$ this is the leading coefficient one. This proves the equality in (4.1); Lemma 2.2 proves the inequality for the actual primitive multiplier. □

Since Q is monic, the primitive multiplier is integral. Thus the local inequalities can be summed without any negative contribution from omitted primes:

$$\log \kappa(Q) \geq 4 \sum_{2M_0 < p < 4M_0} r_p \log p \tag{4.3}$$

for M_0 sufficiently large. This is a lower bound after all coefficient content has been removed.

4.2 Proportional offsets

Take $a_i = m + i$, $M_0 = m + n$, and suppose first that $m/n \to \rho < \infty$. Reverse the finite sum in (4.3). Since the endpoints $2M_0$ and $4a_i$ are even and exceed two, endpoint conventions for primes have no effect. The prime number theorem, uniformly for $2M_0 \leq x \leq 4M_0$, gives

$$\begin{aligned} 4 \sum_{2M_0 < p < 4M_0} r_p \log p &= 4 \sum_{a_i > M_0/2} \bigl(\vartheta(4a_i) - \vartheta(2M_0)\bigr) \\ &= 16 \sum_{a_i > M_0/2} (a_i - M_0/2) + o(n^2) \\ &= \mathcal{A}(\rho) n^2 + o(n^2), \end{aligned} \tag{4.4}$$

where

$$\mathcal{A}(\rho) = \begin{cases} 2(1+\rho)^2, & 0 \leq \rho \leq 1, \\ 8\rho, & \rho \geq 1. \end{cases} \tag{4.5}$$

For example, the Riemann sum giving this formula is $16 \int_0^1 (\rho + t - (\rho+1)/2)_+ \, \mathrm{d}t$. Uniformity of the prime-number-theorem error follows from $\sup_{2M_0 \leq x \leq 4M_0} |\vartheta(x) - x| = o(M_0)$ and $M_0 = O(n)$.

For these nodes the Cauchy product is

$$\det C(a,a) = \frac{\prod_{j=1}^{n-1}(j!)^2}{\prod_{i,j=1}^{n}(2m+i+j)}. \tag{4.6}$$

Stirling's formula, followed by a Riemann sum for the denominator, gives

$$\log\det C(a,a) = -\mathcal{D}(\rho)n^2 + o(n^2), \qquad \mathcal{D}(\rho) = \frac{3}{2} + \int_0^1\int_0^1 \log(2\rho+x+y)\,\mathrm{d}x\,\mathrm{d}y. \tag{4.7}$$

For clarity, $2\sum_{j<n}\log(j!) = n^2\log n - \frac{3}{2}n^2 + O(n\log(n+1))$. At $\rho = 0$ the logarithmic corner singularity is integrable and its Riemann sums converge. Indeed, counting at most k grid points with $i+j=k$ bounds the contribution from $i+j \le \varepsilon n$ by $O(\varepsilon^2(1+|\log\varepsilon|)) + O(\log(n+1)/n)$. A positive offset tending to zero is handled by monotonicity between offsets zero and any fixed $\varepsilon > 0$. Integrating explicitly yields

$$\begin{aligned}\mathcal{D}(\rho) = \frac{1}{2}\big((2\rho+2)^2\log(2\rho+2) - 2(2\rho+1)^2\log(2\rho+1) \\ + (2\rho)^2\log(2\rho)\big),\end{aligned} \tag{4.8}$$

where $0^2\log 0 = 0$.

The derivative of $\mathcal{D}$ is decreasing, and

$$\mathcal{D}'(\rho) = 2\int_0^1\int_0^1 \frac{\mathrm{d}x\,\mathrm{d}y}{2\rho+x+y} \le \mathcal{D}'(0) = 4\log 2 < 4. \tag{4.9}$$

On both pieces of (4.5), $\mathcal{A}'(\rho) \ge 4$. It follows that

$$\mathcal{A}(\rho) - \mathcal{D}(\rho) \ge \mathcal{A}(0) - \mathcal{D}(0) = 2 - 2\log 2. \tag{4.10}$$

Combining (4.3), (4.4), (4.7) and Theorem 3.3 proves the stronger proportional-offset statement

$$\log Z_{n,m}(\theta) \ge (\mathcal{A}(\rho) - \mathcal{D}(\rho))n^2 + o(n^2). \tag{4.11}$$

4.3 Offsets larger than the matrix size

Suppose now that $m/n \to \infty$. All nodes exceed $M_0/2$ eventually. The same reversal of sums gives

$$\begin{aligned}\log H_{n,m} &\ge 16\sum_{i=1}^{n}\big(m+i-(m+n)/2\big) + o(nM_0) \\ &= 8mn + 8n + o(mn).\end{aligned} \tag{4.12}$$

The relative prime-number-theorem error is still uniform on $[2M_0, 4M_0]$, so this argument imposes no upper bound on m. By (4.6) and the numerator estimate just used,

$$\log\det C(a,a) \ge -n^2\big(\tfrac{3}{2} + \log(2(1+m/n))\big) - O(n\log(n+1)) = -o(mn). \tag{4.13}$$

The Cauchy comparison costs only $O(n)$, whence

$$\log Z_{n,m}(\theta) \ge (8 - o(1))mn. \tag{4.14}$$

Every sequence of ratios m_n/n has subsequences converging in the compactified interval $[0,\infty]$. If the liminf asserted in Theorem 1.1 failed, a subsequence witnessing that failure would have a further subsequence with such a limit. Equations (4.11)–(4.14) exclude every possible limit. Positivity follows from Proposition 3.1. This completes the proof of Theorem 1.1.

5 Difference-variable Gram determinants

5.1 Moment formula and target isolation

For a polynomial $P(u,v)$, define the finite integral

$$J(P)=\int_0^1\int_0^1 \frac{P(u,v)\log^2(uv)}{\sqrt{uv}(1+uv)}\,\mathrm{d}u\,\mathrm{d}v, \qquad \mu_m=\frac{1}{96}J((u-v)^{2m}). \tag{5.1}$$

The weight is integrable and strictly positive on the open square.

Lemma 5.1. *With C_m, B_m defined in (1.6)–(1.7),*

$$\mu_m = C_m\theta + B_m. \tag{5.2}$$

Proof. Expand $(1+uv)^{-1}$ as a geometric series. For $a,b>0$, the log-squared integral in each term is the second common-parameter derivative of $((a+t)(b+t))^{-1}$ at $t=0$. For $a\neq b$ this derivative is

$$\frac{2}{b-a}\left(\frac{1}{a^3}-\frac{1}{b^3}\right),$$

and for $a=b$ it is $6/a^4$. The sum of the absolute integrals of the terms is finite (their tails are $O(k^{-4})$), justifying termwise integration. Substitution of $a=k+i+1/2$, $b=k+j+1/2$ gives

$$\frac{J(u^iv^i)}{96}=(-1)^i(\theta-T_4(i)), \tag{5.3}$$

$$\frac{J(u^iv^j)}{96}=-\frac{(-1)^i}{6}\frac{T_3(i)-T_3(j)}{j-i} \quad (i\neq j,\ i+j \text{ even}). \tag{5.4}$$

In (5.4), the two cubic tails have the same sign $(-1)^i=(-1)^j$, so their $\beta(3)$ terms cancel. Expand $(u-v)^{2m}$. Its diagonal term contributes $\binom{2m}{m}(\theta-T_4(m))$ by (5.3). Pairing indices j and $2m-j$ in (5.4) gives exactly the sum in (1.7). This proves (5.2). □

All coefficient statements use the explicitly defined polynomial C_mX+B_m. No independence or irrationality of θ is assumed to identify its coefficients.

5.2 Positive measure and full degree

Under $y=(u-v)^2$, symmetry between the two halves of the square and the Jacobian $\mathrm{d}y=2s\,\mathrm{d}s$, $s=\sqrt{y}$, show that μ_m is the mth moment of a measure on $[0,1]$ with density

$$\rho(y)=\frac{1}{96s}\int_0^{1-s}\frac{\log^2(v(v+s))}{\sqrt{v(v+s)}(1+v(v+s))}\,\mathrm{d}v, \qquad 0<y<1. \tag{5.5}$$

This density is positive throughout $(0,1)$ and has total mass $\mu_0=\theta$. Consequently

$$D_n:=Q_n^{\Delta}(\theta)=\det[\mu_{i+j}]_{0\le i,j<n}>0 \tag{5.6}$$

for every n: a nonzero polynomial has positive squared norm.

Lemma 5.2. *For $n\ge 1$,*

$$\det[C_{i+j}]_{0\le i,j<n}=2^{n-1}. \tag{5.7}$$

Thus Q_n^{Δ} has exact degree n and leading coefficient 2^{n-1}.

Proof. The C_m are the moments of the probability arcsine measure $\mathrm{d}y/(\pi\sqrt{y(4-y)})$ on $[0,4]$. This follows either by the beta integral or by the substitution $y = 2+2\cos t$, $0 < t < \pi$, and the binomial expansion of $(2+2\cos t)^m$. Let T_j denote the Chebyshev polynomial satisfying $\mathsf{T}_j(\cos t) = \cos(jt)$. The monic integer polynomials

$$c_0(y) = 1, \qquad c_j(y) = 2\mathsf{T}_j((y-2)/2) \quad (j \geq 1)$$

are orthogonal for this measure, with squared norms $1, 2, 2, \ldots$. Monicity and integrality follow from $c_1 = y-2$, $c_2 = (y-2)^2 - 2$ and the Chebyshev recurrence for $j \geq 2$. The coefficient change from monomials has determinant one. The moment determinant is therefore the product of these norms, giving (5.7). □

5.3 Matching upper and lower real exponents

Proposition 5.3. *For all $n \geq 1$,*

$$\frac{1}{6144^n n!\, 2^{2n^2+n}} \leq D_n \leq \theta^n 4^{-(n-1)^2}. \tag{5.8}$$

In particular,

$$\log D_n = -(\log 4)n^2 + O(n\log(n+1)). \tag{5.9}$$

Proof. For $j \geq 1$ the monic shifted Chebyshev polynomial on $[0,1]$ has supremum norm $2/4^j$. Change the monomial basis of the Gram matrix to these monic polynomials and apply Hadamard's inequality. Since the measure has total mass θ,

$$D_n \leq \theta^n \prod_{j=1}^{n-1} (2/4^j)^2 = \theta^n 4^{-(n-1)^2}.$$

For the lower bound restrict the integral in (5.5) to $v \in [(1-s)/4, (1-s)/2]$. There $v(v+s) \leq 1/2$, so $-\log(v(v+s)) \geq \log 2 \geq 1/2$, and the denominator of the integrand is at most 2. The integrand is at least $1/8$, giving

$$\rho(y) \geq \frac{1-s}{3072s} \geq \frac{1-y}{6144} \quad (0 < s < 1). \tag{5.10}$$

The second inequality is $2 \geq s + s^2$. Ordering the positive definite Gram matrices therefore gives

$$D_n \geq 6144^{-n} \det\left[\frac{1}{(i+j+1)(i+j+2)}\right]_{0\leq i,j<n} = 6144^{-n} \prod_{j=0}^{n-1} \frac{1}{(2j+2)\binom{2j+1}{j}^2}. \tag{5.11}$$

Here determinant monotonicity follows by conjugating the matrix inequality by the inverse square root of the smaller positive matrix.

We include the norm calculation in the last equality. For weight $1-y$ on $[0,1]$, the monic orthogonal polynomial of degree j is

$$p_j(y) = \frac{(-1)^j (j+1)!}{(2j+1)!(1-y)} \frac{\mathrm{d}^j}{\mathrm{d}y^j}(y^j(1-y)^{j+1}).$$

The numerator is divisible by $1-y$. Integration by parts j times shows orthogonality to degrees below j; every boundary term vanishes. The same calculation with p_j gives its squared norm

$$\frac{j!(j+1)!}{(2j+1)!} \int_0^1 y^j (1-y)^{j+1}\, \mathrm{d}y = \frac{j!^2 (j+1)!^2}{(2j+1)!(2j+2)!} = \frac{1}{(2j+2)\binom{2j+1}{j}^2}.$$

The Gram determinant is the product of these monic norms. Finally $\binom{2j+1}{j} \leq 2^{2j+1}$ in (5.11) gives the lower bound in (5.8). Taking logarithms, using $0 < \theta < 1$ and Stirling's formula, proves (5.9). □

6 Exact primitive arithmetic of the difference Gram

Fix $n \geq 1$ and a prime $p > 4n$. All moment indices used in Q_n^{Δ} satisfy $0 \leq m \leq 2n-2$. In particular $2m < p$, $|j-m| < p$ for the indices in (1.7), and $T_4(m)$ is p-integral. In $T_3(j)$ the only possible odd denominator divisible by p is p itself: the largest odd denominator is at most $8n-9 < 2p$, whereas the next odd multiple is $3p$. It appears when $j \geq (p+1)/2$.

6.1 The first residue and its block

It follows that $p^3 B_m$ is p-integral and

$$p^3 B_m \equiv \frac{\chi_4(p)}{6} \sum_{j=h}^{2m} \binom{2m}{j} \frac{1}{j-m} \pmod{p}, \qquad h = \frac{p+1}{2}. \tag{6.1}$$

An empty sum is zero; in the range of this sum $j > m$, so no zero denominator occurs. Put

$$m_0 = \left\lfloor \frac{p}{4} \right\rfloor + 1, \qquad r = \max\{0, 2n-1-m_0\}. \tag{6.2}$$

For $m < m_0$ no cubic tail contains p, so B_m is actually p-integral. If $m_0 \leq 2n-2$, the first nonzero residue in (6.1), apart from its factor $\chi_4(p)/6$, is

$$\begin{cases} 1/m_0, & p = 4m_0 - 1, \\ (2m_0-1)(m_0+1)/(m_0(m_0-1)), & p = 4m_0 - 3. \end{cases} \tag{6.3}$$

In the first case the sum has only $j = 2m_0$; in the second it has $j = 2m_0 - 1, 2m_0$, giving $2m_0/(m_0-1) + 1/m_0$. These residues are nonzero modulo p. Indeed, in the nonempty case $m_0 \geq 3$, and all the displayed positive numerator and denominator factors are less than p.

The residue matrix $[p^3 B_{i+j} \bmod p]_{i,j<n}$ is supported in the bottom-right $r \times r$ block. That block is anti-triangular with the nonzero first residue on its anti-diagonal, so it has rank r. More strongly, every entry outside it is p-integral before multiplication by p^3: its moment index is below m_0.

6.2 The exact primitive exponent

Proof of (1.9). At most r entries of a permutation term can have negative valuation, because only the bottom r rows have such entries. Each entry has coefficient valuations at least -3, so every coefficient of Q_n^{Δ} has valuation at least $-3r$.

If $r > 0$, let L be the $r \times r$ leading residue block just described, and let $s = n - r$. To survive reduction after multiplication by p^{3r}, a term must use all r bottom rows and all r bottom columns from that block. Its complementary factor is the upper-left pencil. Consequently, as a polynomial congruence,

$$p^{3r} Q_n^{\Delta}(X) \equiv \det(L) \det[C_{i+j} X + B_{i+j}]_{0 \leq i,j < s} \pmod{p}. \tag{6.4}$$

The right-hand coefficient of X^s is $\det(L) 2^{s-1}$ by Lemma 5.2, and is a p-unit. Here $s \geq 1$, since $p > 4n$. Thus the minimum coefficient valuation is exactly $-3r$. If $r = 0$, all entries are p-integral and the leading coefficient 2^{n-1} is already a p-unit. Lemma 2.2 now proves (1.9) in all cases. □

This argument controls the actual primitive multiplier. Any coefficient gcd is already included in the minimum-valuation calculation; it cannot remove the forced exponent.

6.3 Summing the prime cost

By Lemma 2.2 and the leading coefficient 2^{n-1}, $v_p(\kappa_n) \geq 0$ at every odd prime, while $v_2(\kappa_n) \geq -(n-1)$. Hence

$$\log \kappa_n \geq 3 \sum_{4n<p<8n-8} \left(2n - 2 - \left\lfloor \frac{p}{4} \right\rfloor\right) \log p - (n-1)\log 2. \tag{6.5}$$

Changing the endpoints by a bounded amount and removing floors costs $O(n\log(n+1))$, which is negligible on the quadratic scale. Partial summation and (1.11) give

$$\begin{aligned} 3 \sum_{4n<p<8n-8} \left(2n - 2 - \left\lfloor \frac{p}{4} \right\rfloor\right) \log p &= 3 \int_{4n}^{8n} (2n - t/4)\, \mathrm{d}\vartheta(t) + O(n\log(n+1)) \\ &= 3 \int_{4n}^{8n} (2n - t/4)\, \mathrm{d}t + o(n^2) = 6n^2 + o(n^2). \end{aligned} \tag{6.6}$$

For the replacement of $\mathrm{d}\vartheta$ by $\mathrm{d}t$, integration by parts bounds the error using $\sup_{4n\leq t\leq 8n} |\vartheta(t) - t| = o(n)$, the $O(n)$ endpoint weight and its bounded derivative. Thus $\log \kappa_n \geq 6n^2 + o(n^2)$. Combining with Proposition 5.3 proves (1.10) and completes the proof of Theorem 1.2.

7 Scope, evidence and the wider investigation

The two proofs establish actual primitive-value obstructions. The first uses a monic pencil and a unique surviving principal minor; the second uses a power-of-two leading coefficient and a full anti-triangular residue block. In both cases the leading coefficient controls possible negative valuations of the primitive multiplier at the primes not used in the lower-bound sum. Without that control, a large denominator cost alone would not prove the same conclusion.

These results concern the full pencils defined in (1.4) and (1.8). Invertible rational basis changes preserve their primitive polynomials, but different subspaces, sparse designs, nonprincipal minors, polynomial weights, factors and combinations of determinants are outside the two obstruction theorems. The uniform real comparison in Theorem 3.3 does apply to nonprincipal designs; an arithmetic obstruction for such a design would require additional information.

The accompanying research catalog separates proved limitations of defined families from failed estimates, exact reductions to existing constructions, finite experiments and incomplete constructions. In particular, some nonprincipal designs have unfavorable proved denominator bounds without an all-index bound on numerator content. An ordinary-moment first-variation construction has a positive error and a proved real exponent, but no all-index primitive arithmetic conclusion. Neither status is upgraded to an impossibility statement by the present theorems. The catalog and source snapshots provide the basis for a separate technical supplement; they are not substitutes for proofs in this paper.

Computational checks and authorship

This note was prepared with Codex. The archived record includes model mathematical cross-checks, exact rational calculations and Arb interval calculations. A reproducibility record accompanies this draft and states which checks were rerun and how the archived outputs were preserved. Such checks are finite evidence for identities and arithmetic formulas; the asymptotic results are proved in the text. No proof-assistant verification or human mathematical audit is claimed. The comparison with prior work is recorded by source and version; no exhaustive priority claim is made.

A Including a common zero node

The unregularized Cauchy matrix is unsuitable when both node lists contain zero, but the regularized matrix in (3.21) remains finite. For $a_0 = b_0 = 0$ and increasing positive remaining nodes, write a_+, b_+ for the lists with zero removed. Continuity gives $R_\tau(0,0) = 1/\tau$ and

$$\det[R_\tau(a_i, b_j)]_{0\le i,j<n} = \frac{1}{\tau}\prod_{i>0}\frac{a_i}{a_i+\tau}\prod_{j>0}\frac{b_j}{b_j+\tau}\det C(a_+, b_+). \tag{A.1}$$

To prove it, multiply row i by $a_i + \tau$ and column j by $b_j + \tau$. The resulting entry is $\tau + ab/(a+b)$, with continuous value τ in the zero row and column. Subtract the zero row from every other row and expand at the zero column. Undoing the scalings gives (A.1); an empty determinant is one.

The lower density bound (3.18) still applies. For an upper bound finite at zero, note that $1 - \operatorname{sech} v \le v^2/2$ for $v \ge 0$: integrate $\operatorname{sech} v \tanh v \le v$. Therefore

$$q(s) \le 5(1 - \operatorname{sech}(2\pi s)) \le 10\pi^2 s^2, \qquad q(s) \le M.$$

Combining the bounds gives $q(s) \le (M + 10\pi^2\tau^2)s^2/(s^2+\tau^2)$ for every $\tau > 0$. Choose $\tau = \tau_0$; then

$$c_0^n \det R_{\tau_0} \le \det[K(a_i^2, b_j^2)] \le [c_0(M+5)]^n \det R_{\tau_0}. \tag{A.2}$$

For positive integer remaining nodes, (A.1) gives

$$\log\det[K(a_i^2, b_j^2)] = \log\det C(a_+, b_+) + O(n), \tag{A.3}$$

uniformly in the remaining node lists.

Corollary A.1. *For the principal nodes $0, 1, \dots, n-1$, replace θ by X in (3.8)–(3.10), take the determinant, and let Z_n^0 be its primitive representative. Then*

$$\liminf_{n\to\infty}\frac{\log Z_n^0(\theta)}{n^2} \ge 2 - 2\log 2.$$

Proof. The coefficient matrix of X is $\operatorname{diag}(2, 1, \dots, 1)$, so the leading coefficient of the determinant is 2. For primes $2(n-1) < p < 4(n-1)$, the unique high-node minor from Lemma 4.1 is unchanged; its complementary diagonal coefficient contributes a factor 2, a p-unit. Thus the same coefficient valuation $-4r_p$ is attained. All odd prime valuations of the primitive multiplier are nonnegative, and the only possible negative contribution is at least $-\log 2$. The prime sum is $2(n-1)^2 + o(n^2)$. By (A.3), its positive real determinant has logarithm $-2\log 2\,(n-1)^2 + o(n^2)$. Adding proves the assertion. □

References

[1] F. Brown, *Mellin transforms, transfinite diameter and rational approximations of integrals*, preprint, arXiv:2604.20741v1 (2026).

[2] M. C. S. Carreira, *Determinant constructions for the Dirichlet beta value at four*, preprint (2026), pinned PDF in the catalan-designs repository, commit `974a6e6c7b4b` (full identifier in the accompanying source ledger).

[3] C. Krattenthaler, *A determinant identity for moments of orthogonal polynomials that implies Uvarov's formula for the orthogonal polynomials of rationally related densities*, arXiv:2103.03969v1 (2021).

[4] OpenAI, *Catalan's constant is irrational*, manuscript dated 24 September 2026, pinned PDF in the OpenAI math repository, commit `adc7f1241b42` (full identifier in the source ledger).

[5] M. Prévost, *Remainder Padé approximants for the Hurwitz zeta function*, arXiv:1709.05389v1 (2017).

[6] W. Zudilin, *A determinantal approach to irrationality*, arXiv:1507.05697v2 (2015).